
Math Training — M1 Cognitive Science

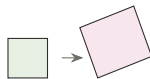
Session 2 — Linear Algebra II

Determinant, rank, eigenvectors, diagonalisation, PCA, SVD

Advanced track

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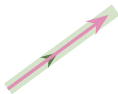
1. **Determinant**, inverse, **rank**
2. **Low rank**: a big table acting small
3. **Eigenvectors**, eigenvalues, diagonalization
4. Symmetric matrices, covariance, **PCA**
5. **SVD**: PCA straight from the table



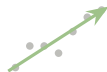
area factor



collapse

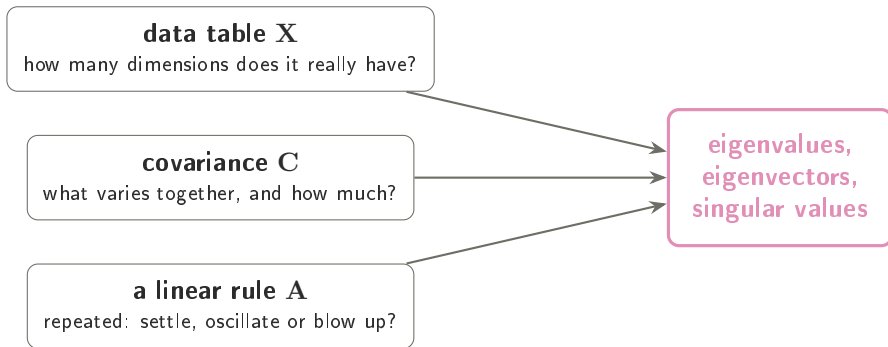


pure stretch



long axis

Three questions, one answer

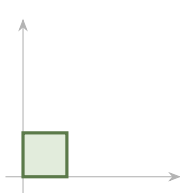


■ Key idea

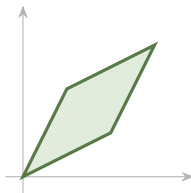
One object, three questions. *Which* matrix depends on the question — the answer is always its spectrum.

The determinant is an area factor

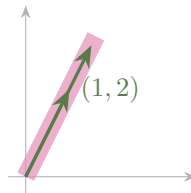
$$\det \mathbf{A} = a_{11}a_{22} - a_{12}a_{21}$$



area 1

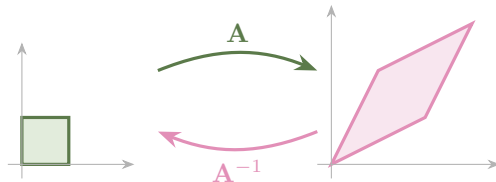


$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$: area 3



$\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$: area 0

$|\det|$ is the area factor; its *sign* says whether orientation is kept — which is why every reflection has a negative determinant.

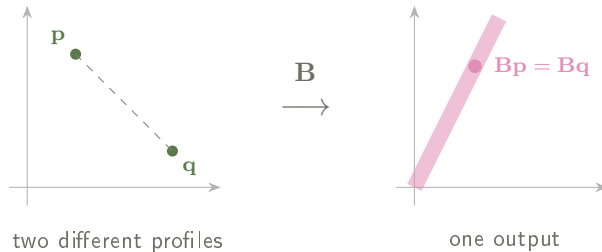


■ Definition

When $\det \mathbf{A} \neq 0$,

$$\mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

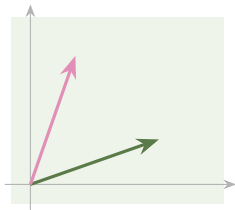
— swap the diagonal, negate the off-diagonal, divide by $\det$.



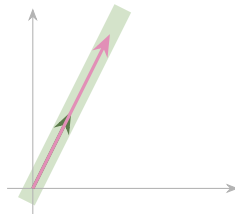
■ Intuition

The map flattens the plane onto a line. Two different inputs give the *same* output — so there is nothing to invert. Information is lost, not hidden.

Rank: the honest count of dimensions



rank 2: outputs fill the plane

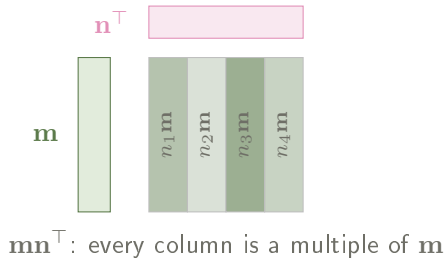


rank 1: outputs on one line

■ Definition

Rank = number of independent columns = dimension of the set of outputs (the column reading of Session 1).

Rank 1 at any size: the outer product



■ Useful fact

$\mathbf{mn}^\top$ has **rank 1** whatever its size. A sum of r outer products has rank $\leq r$, with image in $\text{span}(\mathbf{m}_1, \dots, \mathbf{m}_r)$.

■ Example

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

$$\det \mathbf{A} = 4 - 1 = 3 \neq 0$$

$\Rightarrow$ invertible, **rank 2**

$$\mathbf{A}^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \mathbf{A} \cdot \frac{1}{3} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \checkmark$$

■ Example

$$\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

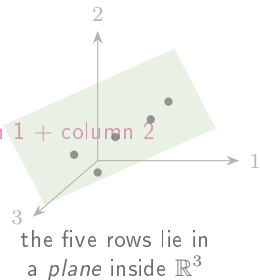
$$\det \mathbf{B} = 4 - 4 = 0$$

$\Rightarrow$ no inverse, **rank 1**

Outputs on the line through $(1, 2)$, and
 $\mathbf{B} = \mathbf{m}\mathbf{n}^\top$ with $\mathbf{m} = \mathbf{n} = (1, 2)$.

	item 1	item 2	total
P_1	4	9	13
P_2	6	7	13
P_3	5	8	13
P_4	9	4	13
P_5	7	6	13

column 3 = column 1 + column 2
 $\Rightarrow$ rank 2, not 3



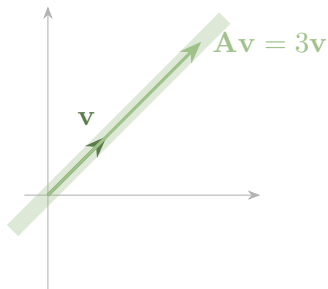
■ Link to cognitive science

Include a total score *alongside* the items it is made of and the table is rank-deficient by construction. In a regression on those three predictors, $\mathbf{X}^\top \mathbf{X}$ is singular: **infinitely many weight vectors give exactly the same predictions**, so no single “effect of item 1” exists. This is collinearity.

■ Practice — try it

1. Compute $\det\begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$ and its inverse.
2. Rank of $\begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix}$? Invertible? On what line do its outputs lie?
3. For which k is $\begin{pmatrix} 1 & 2 \\ 2 & k \end{pmatrix}$ non-invertible? What happens to the rank there?

The directions that survive

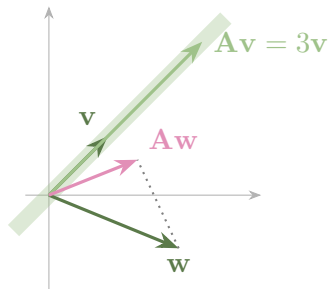


■ Definition

$\mathbf{v} \neq \mathbf{0}$ is an **eigenvector** if $A\mathbf{v} = \lambda\mathbf{v}$. The stretch factor λ is the **eigenvalue**.

$\mathbf{v}$ stays on the green line, three times longer. $\mathbf{w}$ is knocked off its own line entirely.

The directions that survive



■ Definition

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Finding them: one quadratic

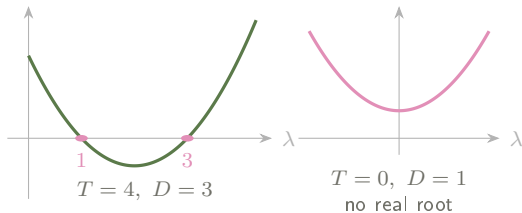
$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

$$\lambda^2 - T\lambda + D = 0$$

$$T = \text{tr } \mathbf{A}, \quad D = \det \mathbf{A}$$

■ Two free checks

$$\lambda_1 + \lambda_2 = T, \quad \lambda_1 \lambda_2 = D.$$



Roots of the parabola. No crossing $\Rightarrow$ complex pair.

■ Example

$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$: $T = 4, D = 3 \Rightarrow \lambda^2 - 4\lambda + 3 = 0 \Rightarrow \lambda = 1, 3$. Check: $1 + 3 = 4$
 $\checkmark, 1 \cdot 3 = 3 \checkmark$

$\lambda = 3$:

$$(\mathbf{A} - 3\mathbf{I})\mathbf{v} = \mathbf{0} \Rightarrow -v_1 + v_2 = 0$$

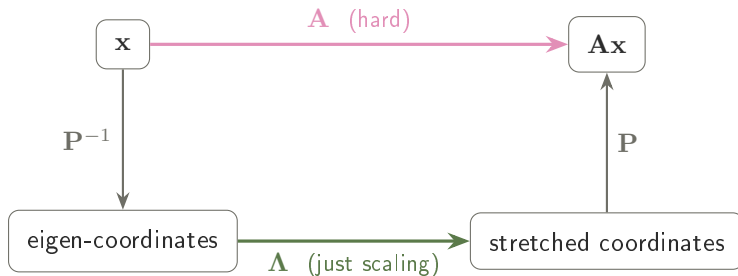
$$\mathbf{v} = (1, 1)$$

$\lambda = 1$:

$$(\mathbf{A} - \mathbf{I})\mathbf{v} = \mathbf{0} \Rightarrow v_1 + v_2 = 0$$

$$\mathbf{v} = (1, -1)$$

$(1, 1) \cdot (1, -1) = 0$: orthogonal. Not an accident — see the spectral theorem.

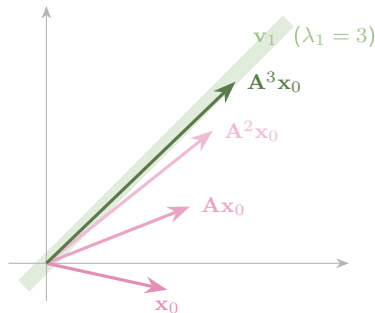


■ Useful fact

Eigenvectors as the columns of $\mathbf{P}$, eigenvalues along $\mathbf{\Lambda}$, same order:

$$\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1} \quad \implies \quad \mathbf{A}^k = \mathbf{P}\mathbf{\Lambda}^k\mathbf{P}^{-1}.$$

Repeat it enough and the largest $|\lambda|$ wins



■ Key idea

$A^k \mathbf{x}_0 = c_1 \lambda_1^k \mathbf{v}_1 + c_2 \lambda_2^k \mathbf{v}_2$. The ratio $(\lambda_2/\lambda_1)^k \rightarrow 0$, so the direction converges to $\mathbf{v}_1$.

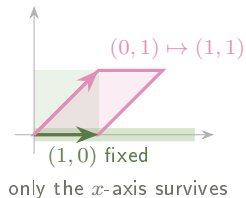
Provided $|\lambda_1|$ is *strictly* largest and $c_1 \neq 0$ — true for a random start.

■ Watch out

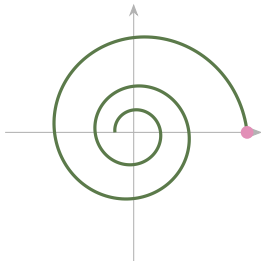
$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has $\lambda = 1$ *twice* but only *one* eigendirection, $(1, 0)$.

Such matrices are **defective**: no eigenbasis, no diagonalization.

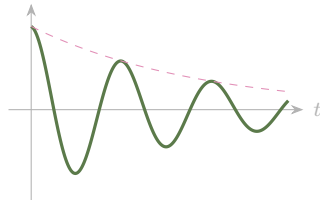
Two *distinct* eigenvalues always suffice. A repeated one may or may not.



Complex eigenvalues mean rotation



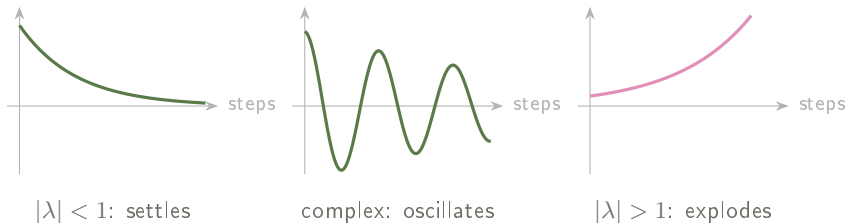
$\mu < 0$: spiral inward



amplitude $e^{\mu t}$, rhythm ω

■ Key idea

$\lambda = \mu \pm i\omega$ (always a conjugate pair for a real matrix): ω is the angular speed, μ decides growth or decay.



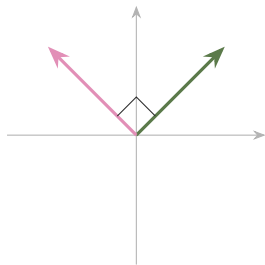
■ Link to cognitive science

Repeat a linear rule $\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k$ and the eigenvalues decide which of these three you get — nothing else does. **Do not repeat anything**, and eigenvalues still matter: for a *covariance* matrix they are simply the variances along the principal directions. Same object, two very different uses.

■ Practice — try it

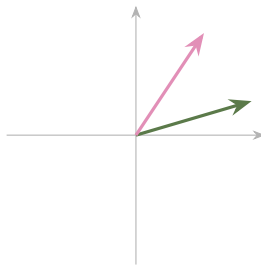
1. Verify $(1, 0)$ is an eigenvector of $\begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}$; give its eigenvalue.
2. For $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ compute T and D , then the eigenvalues.
3. A matrix has eigenvalues 2 and 0.5. Apply it many times to a generic vector: which direction does the result line up with, and why?

Symmetric matrices are especially nice



$$\mathbf{A} = \mathbf{A}^T$$

eigenvectors **perpendicular**

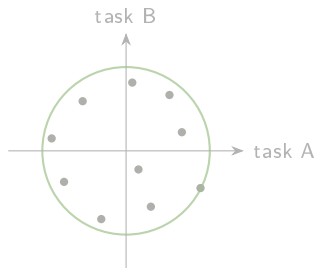


$$\mathbf{A} \neq \mathbf{A}^T$$

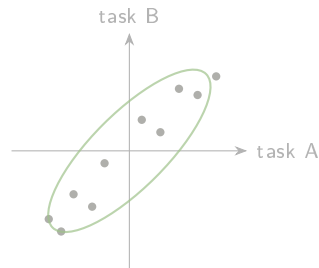
eigenvectors **oblique**

■ Spectral theorem

A *real symmetric* matrix has *real* eigenvalues and *orthogonal* eigenvectors — so its eigenbasis is a rotation, and $\mathbf{P}^{-1} = \mathbf{P}^T$.



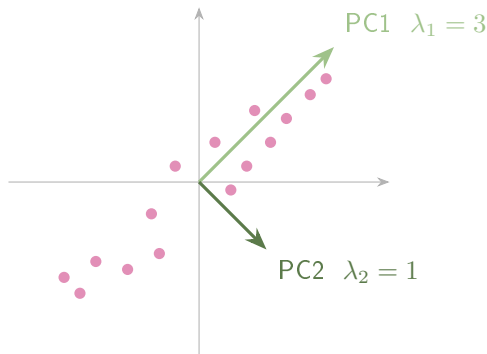
$\begin{pmatrix} v & 0 \\ 0 & v \end{pmatrix}$: uncorrelated



$\begin{pmatrix} v & c \\ c & v \end{pmatrix}$: correlated, tilted

■ Definition

For centred data, $\mathbf{C} = \frac{1}{n} \mathbf{X}^\top \mathbf{X}$: variances on the diagonal, co-variation off it. Symmetric, and **positive semi-definite**, since $\mathbf{u}^\top \mathbf{C} \mathbf{u}$ is the variance of the data projected on $\mathbf{u}$, hence ≥ 0 .



■ Key idea

Principal components = **eigenvectors of $\mathbf{C}$** . Each eigenvalue = **variance along that direction**. Here PC1 carries $3/4 = 75\%$ of the spread.

■ Example

Two similar tasks, centred scores, $\mathbf{C} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ — the same matrix again.

$\lambda_1 = 3$ along $(1, 1)$: **overall level**

$\lambda_2 = 1$ along $(1, -1)$: **B – A**

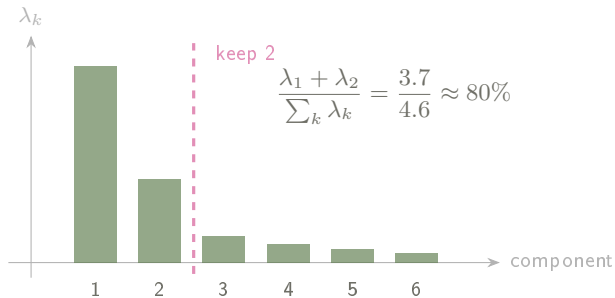
Both real, both ≥ 0 , and orthogonal —
exactly as the spectral theorem and positive
semi-definiteness promise.

$$\frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{3}{4} = 75\%$$

along $(1, 1)$; the remaining 25% along
 $(1, -1)$.

The rotation we *guessed* in Session 1 is
the one PCA *finds*.

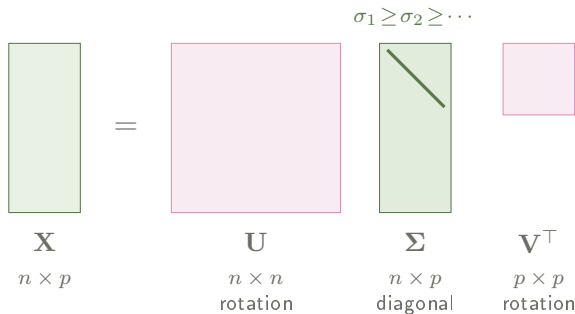
How many components?



■ Link to cognitive science

A battery of six tests, but the scores really move along *two* directions. Two honest criteria: the **cumulative variance explained**, and the **elbow** in the plot. Neither is a proof — “how many components” is a judgement, and it should be reported as one.

SVD: straight from the table

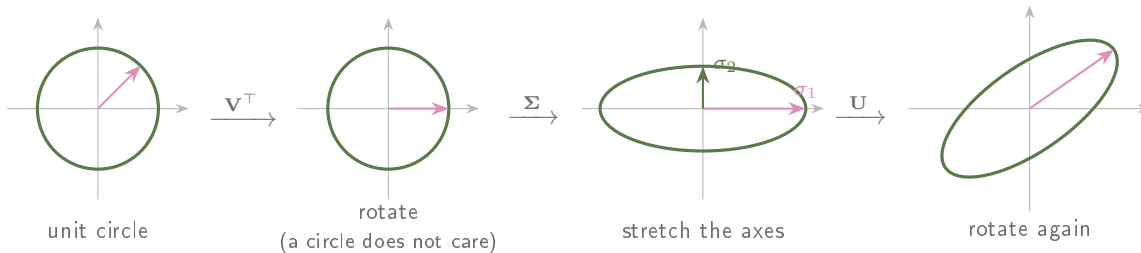


■ Key idea

$$\mathbf{X} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$$

Every matrix factors this way — any shape, any rank, no exceptions. $\mathbf{U}$ and $\mathbf{V}$ are orthogonal ($\mathbf{U}^T \mathbf{U} = \mathbf{I}$); $\mathbf{\Sigma}$ is diagonal with $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$.

What the three factors do: rotate, stretch, rotate



■ Key idea

Every linear map is **a rotation, a stretch along perpendicular axes, and another rotation**. The σ_k are the semi-axes of the ellipse: how far the map reaches in each direction.

$$\mathbf{X}^\top \mathbf{X} = (\mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top)^\top (\mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top) = \mathbf{V} \mathbf{\Sigma}^\top \underbrace{\mathbf{U}^\top \mathbf{U}}_{=\mathbf{I}} \mathbf{\Sigma} \mathbf{V}^\top = \mathbf{V} \mathbf{\Sigma}^2 \mathbf{V}^\top$$

■ Useful fact

For centred data, $\mathbf{C} = \frac{1}{n} \mathbf{X}^\top \mathbf{X}$, so

$$\underbrace{\text{columns of } \mathbf{V}}_{\text{right singular vectors}} = \underbrace{\text{eigenvectors of } \mathbf{C}}_{\text{principal components}}, \quad \lambda_k = \frac{\sigma_k^2}{n}.$$

PCA is the SVD of the centred data table. Variance explained: $\sigma_k^2 / \sum_j \sigma_j^2$.

$$\mathbf{X} = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^\top + \cdots + \sigma_r \mathbf{u}_r \mathbf{v}_r^\top$$

keep these throw the rest away

■ Eckart–Young

Among *all* matrices of rank r , the truncation $\mathbf{X}_r = \sum_{k \leq r} \sigma_k \mathbf{u}_k \mathbf{v}_k^\top$ is the closest one to $\mathbf{X}$.

■ Link to cognitive science

A table is written as a **sum of outer products**, biggest first — and each term is a rank-1 “one profile, one level per person” pattern from earlier today. Twenty questionnaire items become two component scores; a words $\times$ documents table truncated to a few dimensions gives directions that behave like topics; and the terms you dropped, the small σ_k , are where the noise was living.

■ Example

Four participants, two measures, already centred: $\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ -1 & -1 \\ -1 & -1 \end{pmatrix}$

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix} : \quad T = 8, \quad D = 0$$

$$\Rightarrow \sigma_1^2 = 8, \quad \sigma_2^2 = 0 \Rightarrow \sigma_1 = 2\sqrt{2}, \quad \sigma_2 = 0$$

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1), \quad \mathbf{u}_1 = \frac{\mathbf{X}\mathbf{v}_1}{\sigma_1} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}$$

Rank 1: $\mathbf{X} = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top$ exactly.

Variance explained by PC1:

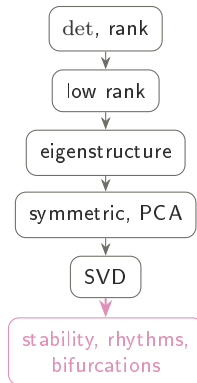
$$\frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} = \frac{8}{8} = 100\%.$$

One profile ($\mathbf{v}_1$: both measures together)
and one number per participant reproduce
the whole table.

■ Practice — try it

1. $\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$: symmetric? Find its eigenvalues and check the eigenvectors are orthogonal.
2. A covariance matrix has eigenvalues 5 and 0.2. Along which eigenvector does the data mostly spread, and what fraction of the variance does it carry?
3. A centred table $\mathbf{X}$ has singular values $\sigma = (4, 3, 1)$. Give the variance explained by the first two components, and the rank of the best rank-2 approximation's error.
4. True or false: a real symmetric matrix can have complex eigenvalues. Justify.

- **Determinant** = signed area factor; zero = no inverse. **Rank** = true dimensionality.
- **Outer products**: rank 1 at any size — one profile, one level per person.
- **Eigenvectors** = directions only stretched; $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}$ makes powers trivial; complex = rotation.
- **Real symmetric** $\Rightarrow$ real, orthogonal — which is what makes **PCA** work.
- **SVD** $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$: PCA without ever forming $\mathbf{C}$, and the best low-rank summary of any table.



Next: differential equations — what it means to *solve* one, and why the exponential is the only shape that works.